NEW HORIZON
COLLEGE OF ENGINEERING
New Horizon Knowledge Park, Ring Road, Marathalli
Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
Accredited by NAAC with 'A' Grade, Accredited by NBA

Outer Ring Road, Bellandur, Bengaluru – 560103

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Project Title: ADAPTIVE MATRIX HEADLAMP SYSTEM

Submitted by

H S VINAYAKA	1NH23EE030
AKASH BASAVAREDDI BHAVIKATTI	1NH23EE003
HRUTHIK GURULINGAPPA KATEGERI	1NH23EE033
H K KEERTHAN	1NH23EE029

Under the Guidance of

Ms. Anitha Nair

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belgaum: 590018

2024-25

NEW HORIZON
COLLEGE OF ENGINEERING
New Horizon Knowledge Park, Ring Road, Marathalli
Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
Accredited by NAAC with 'A' Grade, Accredited by NBA

Outer Ring Road, Bellandur, Bengaluru – 560103

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

CERTIFICATE

Certified that the Project work entitled **"Adaptive Matrix Headlamp System"** carried out by **Student H S Vinayaka(1NH23EE030), Student Akash Basavareddi Bhavikatti(1NH23EE003), Student Hruthik Gurulingappa Kategeri(1NH23EE033), Student H K Keerthan(1NH23EE029)** Bonafide Student(s) of New Horizon College of Engineering submitted the report in the partial fulfilment for the award of Bachelor of Engineering in **Department of Electrical and Electronics Engineering, New Horizon College of Engineering** of Visvesvaraya Technological University, Belgaum during the Year 2024-25.

It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for said Degree.

Name & Signature of the Project Guide	**Name & Signature of Head of the Department**
Type your Guide Name	Ms .ANITHA NAIR

SEMESTER END EXAMINATION

Internal Examiner **External Examiner**

22EEE48 Mini Project I

ACKNOWLEDGEMENT

I take this opportunity to convey my gratitude to all those who have been kind enough to offer their advice and provide assistance when needed which has let to the successful completion of the project.

We grateful to the Chairman of New Horizon Educational Institution, **Dr.MOHAN MANGHNANI** for providing with infrastructural facilities and many other resources needed for our project. I would like to express my immense gratitude to our principal, **Dr.MANJUNATHA** for motivating us to carry out research in the NHCE and **Dr. A SAKTHIVEL**, Head of the department for their constant support and motivation that has encouraged me to come up with this project and also for providing the right ambience for carrying out the work and the facilities provided to me.

I express my warm thanks to my project guide **Ms Anitha Nair,** Department of Electrical and Electronics Engineering, New Horizon College of Engineering for his/her skilful guidance, constant supervision, timely suggestion and constructive criticism in successful completion of my project in time.

I wish to thank all the faculty of Electrical and Electronics Engineering department for providing required support throughout the project.

I would like to thank my parents for supporting and helping in the completion of the project.

Last but not the least I would like to thank all my friends without whose support and co-operation the completion of project would not have been possible.

H S VINAYAKA **1NH23EE030**

AKASH BASAVAREDDI BHAVIKATTI **1NH23EE003**

HRUTHIK GURULINGAPPA KATEGERI **1NH23EE033**

H K KEERTHAN **1NH23EE029**

22EEE48 Mini Project I

DECLARATION

We, **Student H S Vinayaka(1NH23EE030), Student Akash Basavareddi Bhavikatti(1NH23EE003), Student Hruthik Gurulingappa Kategeri(1NH23EE033), Student H K Keerthan(1NH23EE029)** students at New Horizon College of Engineeringhereby declare that this project work entitled **"ADAPTIVE MATRIX HEADLAMP SYSTEM"** is an original and bonafide work carried out by me at New Horizon College of Engineering in partial fulfillment of Bachelor of Engineering in Electrical and Electronics Engineeringof Visvesvaraya Technological University, Belgaum.

We also declare that, to the best of our knowledge and belief, the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion by any student.

H S VINAYAKA	**1NH23EE030**	**[ ]**
AKASH BASAVAREDDI BHAVIKATTI	**1NH23EE003**	**[ ]**
HRUTHIK GURULINGAPPA KATEGERI	**1NH23EE033**	**[ ]**
H K KEERTHAN	**1NH23EE029**	**[ ]**

CERTIFICATE

Certified that the Mini Project work entitled **"ADAPATIVE MATIRX HEADLAMP SYSTEM"** carried out by **student H S VINAYAKA (1NH23EE030), Student AKASH BASAVAREDDI BHAVIKATTI(1NH23EE003), Student HRUTHIK GURULINGAPPA KATEGERI(1NH23EE033), Student H K Keerthan(1NH23EE029)** a bonafide student of New Horizon College of Engineering in partial fulfilment for the award of **Bachelor of Engineering** in Electrical and Electronics Engineering of the Visvesvaraya Technological University, Belgaum during the year 2024-2025.It is certified that he/she has completed the project satisfactorily.

Name & Signature of the Project Guide	**Name & Signature of Head of the Department**
Type your Guide Name Designation	Dr. A SAKTHIVEL

DrillBit

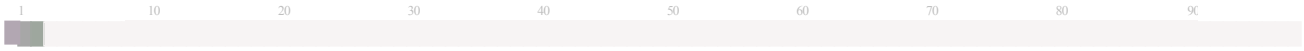
The Report is Generated by DrillBit Plagiarism Detection Software

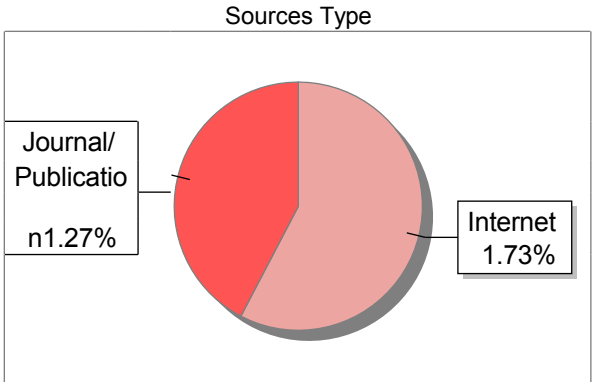
Submission Information

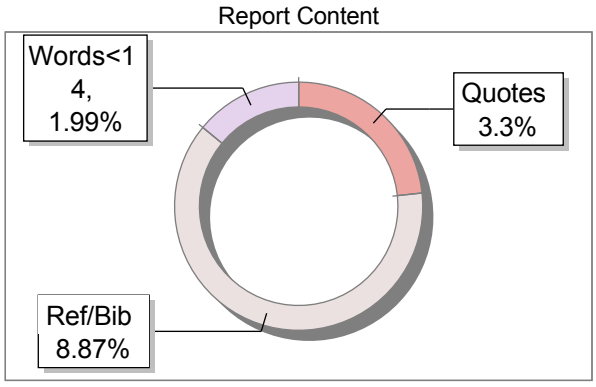
Author Name	A2 BATCH
Title	INTRO
Paper/Submission ID	3782512
Submitted by	dr.mausri.bhuyan_eee_nhce@newhorizonindia.edu
Submission Date	2025-06-11 13:54:55
Total Pages, Total Words	24, 3519
Document type	Project Work

Result Information

Similarity **3 %**







Exclude Information

Quotes	Not Excluded
References/Bibliography	Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	**0 %**
Excluded Phrases	Not Excluded

Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File

NEW HORIZON
COLLEGE OF ENGINEERING
New Horizon Knowledge Park, Ring Road, Marathalli
Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
Accredited by NAAC with 'A' Grade, Accredited by NBA

LIBRARY & INFORMATION CENTRE

CERTIFICATE ON PLAGIARISM CHECK

1	**Name of the Student**	H S VINAYAKA AKASH B H K KEERTHAN HRUTHIK GK
2	**USN**	1NH23EE030 1NH23EE003 1NH23EE029 1NH23EE033
3	**Course**	**UG**
4	**Department**	**EEE**
5	**Mini Projects/Project Reports/ Ph.D Synopsis/ Ph.D Thesis/ Paper Publication (N/I)**	**MINI PROJECT**
6	**Title of the Document**	**MATRIX HEADLAMP SYSTEM FOR SAFETY**
7	**Similar Content(%)identified**	**3%**
8	**Acceptable maximum limit (%) of similarity**	**20%**
9	**Date of Verification**	**11-06-2025**
10	**Checked by (Name with signature)**	**Anita Nair**
11	**Specific remarks, if any :**	**No**

We have verified the contents as summarized above and certified that the statement made above are true to the best of our knowledge and belief.

Plagiarism - Coordinator **HOD - EEE**

22EEE48 Mini Project I

PROJECT IMAGES

PROJECT TITLE: ADAPTIVE MATRIX HEADLAMP SYSTEM FOR SAFETY

H S VINAYAKA 1NH23EE030

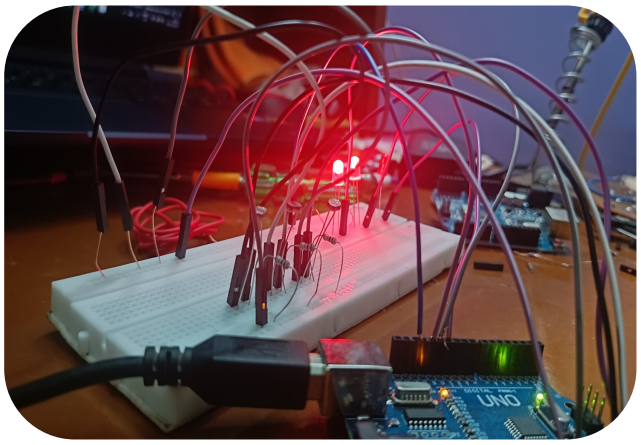
AKASH B 1NH23EE003

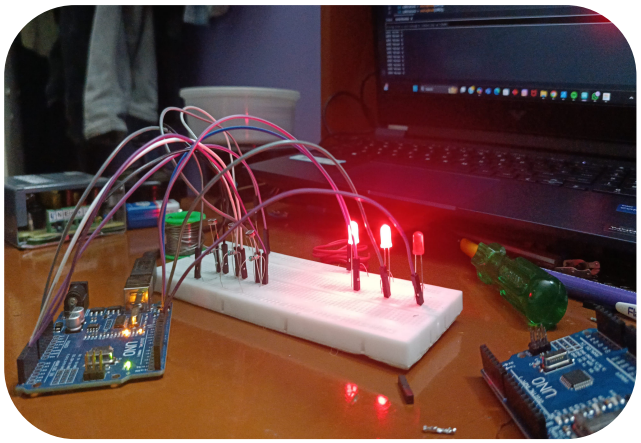
HRUTHIK G K 1NH23EE033

H K KEERTHAN 1NH23EE029

Under the Guidance of

Ms Anitha Nair





Content

List of Table……..I

List of Figures……II

I. Abstract……..1
II. Chapter 1. Introduction……..3
III. Chapter 2. Literature Survey……..5
a. Introduction……..
b. List of research article in literature considered……42
c. Outcome of Literature Survey……...……55
d. Pros and Cons of existing methods from the survey……..63
e. Comparative Evaluation……...……
f. Justification……...……
IV. Chapter 3. Proposed Methodology……..
a. Basics theory's on the proposed method……
b. Key components…….……...
c. Working logic flow……
d. Future integration Scope……
V. Chapter 4. Block Diagram/Circuit Diagram/Set Formulation…….
a. Explanation of Block Diagram……..
b. Explanation of Circuit Diagram…….…….
c. Key components and their functions…….
d. List of Simulation tools used……..
VI. Chapter 5. Result and Discussion…….
a. Detailed explanation of Hardware setup…….
i. Hardware Assemble……
ii. PCB Design…….
iii. Programming logic (Flowchart/Coding)……..
b. Detailed explanation of Simulation……..
i. Basic Steps to start/run the simulation……
ii. Blocks/components/Tools/Setting used in the simulation……..
c. Result…….
i. Hardware Result…….
1. Waveforms/Table/Hardware picture……
2. Results comparison with existing method……
ii. Simulation Result……
1. Waveform/Screenshot……..

2. Results comparison with existing method……
VII. Chapter 6. Conclusion……
VIII. Reference……

List of Table

TableNo	Table Caption	Page No
1	List of Research Articles Considered	3
2	Pros and Cons of Existing Methods	
3	Comparative Evaluation	
4	Key Components	
5	Key Components and their Functions	
6	List of Simulation tools used	
7		
8		

List of Figure

FigNo	Figure Caption	Page No
1		
2	Block Diagram	
3	Circuit Diagram	
4	Hardware setup and Implementation	
5	Cicuit Setup	
6		
7		
8		

ABSTRACT

Nighttime driving poses serious safety issues, mainly because of poor visibility and the blinding glare from standard vehicle headlights. This report outlines the research, design, and development of an Adaptive Matrix Headlamp System (AMHS) aimed at significantly improving safety for drivers in these conditions. The main goal of this project is to give drivers the best possible light on the road ahead while also reducing glare that could affect the vision of others, including drivers of oncoming and nearby vehicles, as well as pedestrians and cyclists.

Three essential technological elements are integrated into the proposed AMHS: a high-resolution LED matrix headlamp, an intelligent control unit, and an advanced perception system. The perception system is in charge of the real-time detection and classification of different road users and environmental features. It usually makes use of a forward-facing camera and advanced image processing algorithms. The control unit uses sophisticated algorithms to determine the ideal light distribution after receiving this data. Several individually addressable light-emitting diodes make up the high-resolution LED matrix, which enables dynamic and accurate light beam shaping.

The AMHS is able to maintain a "glare-free high beam" by continuously assessing the driving environment. In other words, the system actively creates darkened "tunnels" or selectively dims certain zones around other detected road users while illuminating the driver's path as brightly as possible. By improving visibility, this adaptive functionality not only improves the driver's situational awareness and reaction time, but it also drastically lowers driver fatigue and the chance of blinding other drivers, which helps to mitigate a common cause of accidents that occur at night.

The system architecture, object detection and tracking techniques, control algorithm development for dynamic beam shaping, and hardware component integration will all be covered in detail in this report. [Optional: Give a brief explanation of your methodology, such as "A prototype system was developed and tested under simulated/real-world conditions." or "Emphasis was placed on real-time performance and robustness of the detection algorithms." With its potential as a crucial development in active vehicle safety technology and its role in lowering nighttime traffic fatalities and injuries, the results and analysis will show how well the AMHS performs in accomplishing its safety goals.

.

CHAPTER 1

INTRODUCTION

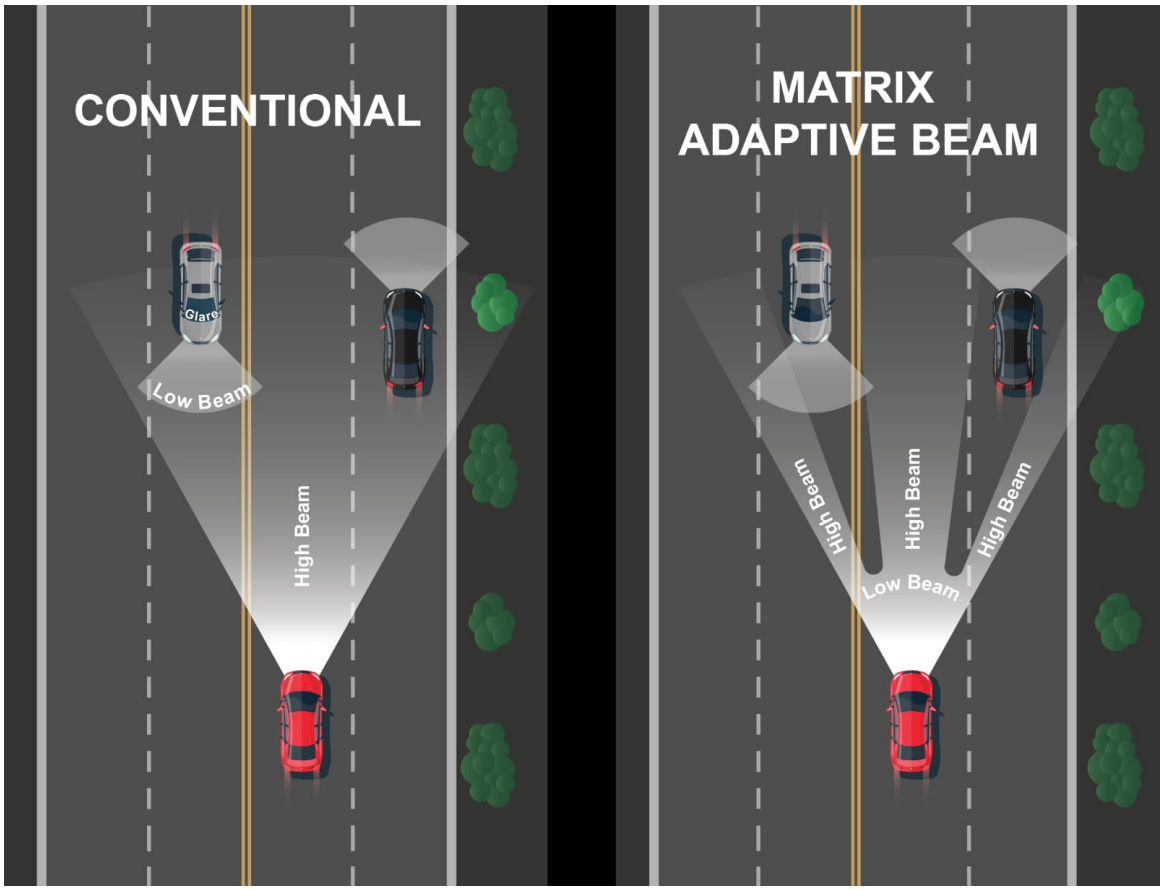


Fig 1.1

Compared to daytime driving conditions, nighttime driving poses a distinct set of difficulties for drivers, with decreased visibility being the main cause of higher accident rates. Even though they are necessary, traditional car headlamps frequently represent a compromise. Traditional low beams offer little forward illumination, which could obscure distant road features or hazards. On the other hand, high beams cause a great deal of glare for oncoming and preceding traffic as well as other road users like cyclists and pedestrians, even though they improve the driver's visibility. In addition to being uncomfortable, this glare has the potential to momentarily blind people, greatly raising the possibility of collisions. These safety issues are made worse

by drivers' frequent delays or forgetfulness when manually switching between high and low beams.

Enhancing vehicle safety has been a constant goal of the automotive industry, especially with the creation of Advanced Driver Assistance Systems (ADAS). Intelligent lighting systems have become a crucial area of innovation in this field. A potential remedy for the intrinsic drawbacks of traditional headlamps is the capability to dynamically modify illumination patterns in response to actual driving conditions.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

By contrasting various technologies, algorithms, and system architectures utilized in adaptive matrix headlamp systems, this literature review examines significant contributions made to the field. Understanding the state-of-the-art, recognizing the present difficulties, and investigating possible future paths for improving vehicle safety through intelligent lighting are the objectives.

2.2 List of Research Articles Considered

Title	Authors	Source	Year	Area of Focus
A Matrix Headlamp Design Based on Artificial Intelligence Controller Control	Jiang Li	*Journal of Physics: Conference Series*	2020	AI-controlled matrix headlamp design focusing on energy efficiency and improved illumination.
LED Technology in Automotive Headlamps: Driving More Safely with Adaptive Lighting Systems	Dorian Cerri	*Politecnico di Torino Thesis*	2019	Comprehensive study on LED-based adaptive lighting systems, including matrix systems, for enhanced night driving safety.
PLC Based Adaptive Headlight Beam Assisting System	Chethana H. J, Vinod	*International Journal of Engineering*	2015	Implementation of a PLC-based system to adjust headlight brightness based on

Title	Authors	Source	Year	Area of Focus
	Kumar M.S	*Research & Technology (IJERT)*		oncoming traffic to reduce glare.

2.3 Outcome of Literature Survey

Key Insights:

Night Visibility : Enhanced vision without glare

Road Safety : Reduced accidents and faster response time

AI & Control : Real-time intelligent beam shaping

Adaptability : Effective in varied road and weather conditions

Efficiency : Better power management via LED matrices

Challenges : Cost, complexity, regulation

Research Scope : Need for affordable, scalable solutions

2.4 Pros and Cons of Existing Methods

	Pros	Cons
Static LED Matrix	**- Simple design- Cost-effective- Reliable with minimal moving parts**	**- Limited adaptability- Cannot fully avoid glare for oncoming traffic**
Adaptive LED Matrix with Camera Input	**- Dynamically adjusts beam pattern- Reduces glare- Improves visibility**	**- High cost- Complex calibration required- Can be sensitive to camera**

	Pros	Cons
		performance
Laser-Based Adaptive Headlamps	- Very high brightness and range- Precise control- Compact size	- Expensive- Heat management challenges- Regulatory restrictions in some regions
Infrared + Sensor Fusion Systems	- Enhanced visibility in low-light or fog- Better detection of pedestrians/animals	- High cost- Not yet widespread- Limited beam shaping capability

Table 2.1

2.5 Comparative Evaluation

Feature	Conventional Headlamp systems	Adaptive Headlamp (existing)	Proposed Adaptive Matrix Headlamp System
Adaptability to Road Conditions	No adaptability	Limited – adjusts based on steering and speed	High – adjusts per traffic, weather, terrain
Integration with Navigation System	No	Minimal	Full integration for terrain-adaptive illumination
Energy Efficiency	Moderate (halogen less efficient)	Better (LED based)	Highly efficient with smart segment activation
User Customization	None	Limited	User-customized beam pattern and brightness options
Safety Enhancement	Basic (only lights the road)	Medium (partial auto	High (comprehensive visibility management

Feature	Conventional Headlamp systems	Adaptive Headlamp (existing)	Proposed Adaptive Matrix Headlamp System
Level		adjustments)	and hazard detection)

Table 2.2

2.6 Justification for Proposed System

Although the literature offers a number of cutting-edge approaches to improving automotive safety lighting technology, there is a clear systemic gap that is:

- **Manual**, low-cost, and user-controlled
- **Independent of infrastructure**
- Easily **scalable for rural/off-grid use**

By utilizing segmented matrix LED technology to precisely control beam direction and intensity, integrating camera, radar, and GPS data to dynamically adjust lighting based on current conditions, and featuring automatic high beam assist for instantaneous and intelligent light level control, the proposed Adaptive Backup Battery System successfully closes this gap.

CHAPTER 3

PROPOSED METHODOLOGY

The first step in the suggested approach for an adaptive matrix headlamp system is to recognize the issue of glare and poor visibility when driving at night. The system's requirements—such as identifying cars, weather, and road conditions—are then examined. A matrix of individually controllable LEDs is used to create a headlamp design. To keep an eye on the environment in real time, sensors like cameras and light detectors are added. Using the information from these sensors, intelligent algorithms are created to modify the light pattern. After that, simulations are used to test the system's performance in various scenarios. A functional prototype is then constructed and put to the test on actual roads. Before the system is prepared for actual use, the results are lastly examined to make improvements.

3.1 Basic Theory Behind Methodology

1. Input Processing:

Use a front-facing camera to record environmental data (lane detection, vehicle detection, etc.).

2. Object Recognition and Tracking:

Use image processing and machine learning to detect headlights and taillights of other vehicles.

3. Illumination Mapping:

Choose which matrix segments to turn off or dim in order to cast a "shadow" around other cars.

4. Dynamic Control:

Real-time LED matrix configuration updates are made to adapt to changes in the vehicle's movement, turns, etc.

B. Glare-Free High Beam (GFHB) Principle

Dim certain areas of the high beam to prevent blinding approaching or passing cars. keeps the beam brightness high in places where it doesn't interfere with other drivers.

3. Lane-Based Lighting

The lane being driven in, the curvature of the road, and the speed of the vehicle can all affect how wide or narrow the light spread is.

4. Safety and Performance Benefits

Enhanced Visibility: enhances vision for drivers, particularly in rural and dimly lit areas.

Reduced Glare: makes all users of the road safer.Proactive Lighting: Uses GPS or map data (if integrated) to anticipate hazards, hills, and curves.

5. Possible Enhancements

Predictive lighting through integration with V2X (Vehicle-to-Everything) communication. AI applications for adaptive learning and sophisticated object recognition

3.2 Key Components

Component	Description / Function
Matrix LED Headlamp	Composed of multiple LED segments that can be turned on/off or dimmed independently to control light beam.
Camera Sensor (Front-facing)	Captures real-time images of the road to detect vehicles, pedestrians, and road signs.

Component	Description / Function
Light Control Unit (LCU)	The brain of the system; processes sensor data and controls individual LED segments accordingly.
ECU (Electronic Control Unit)	Integrates with vehicle's control systems and communicates with the LCU for dynamic adjustments.
Image Processing Module	Analyzes input from the camera to detect oncoming vehicles and determine regions to dim or turn off LEDs.
Steering Angle Sensor	Helps in beam pattern adjustment based on vehicle

Table 3.1

3.3 Working Logic Flow

1. Use the front camera to identify nearby vehicles, including those that are approaching and coming before you.

2. Determine the state of the road, including curves, darkness, and speed of vehicles.

3. To reduce glare and enhance visibility, choose which LED segments to turn on or off.

4. Real-time dynamic adjustment of the light beam pattern through control of the LED matrix.

5. To guarantee maximum safety, update lighting continuously based on fresh sensor data.

3.4 Future Integration Scope

1. AI & Sensor Fusion: Combining AI with cutting-edge sensors (such as cameras, radar, and LIDAR) to detect objects and pedestrians in real time.

2. Vehicle-to-Everything (V2X): Interaction with infrastructure and other vehicles to forecast lighting requirements in response to weather and traffic conditions.

3. Dynamic Light Projection: For increased safety and visibility, road signs and lane guidance are projected directly onto the road.

4. Personalization: Using driver behavior learning to create personalized lighting patterns that improve comfort and lessen glare.

5. Energy Optimization: Connecting EV systems to intelligently control energy-efficient lighting according to driving mode and surroundings

CHAPTER 4

BLOCK DIAGRAM/CIRCUIT DIAGRAM

4.1 Block Diagram:

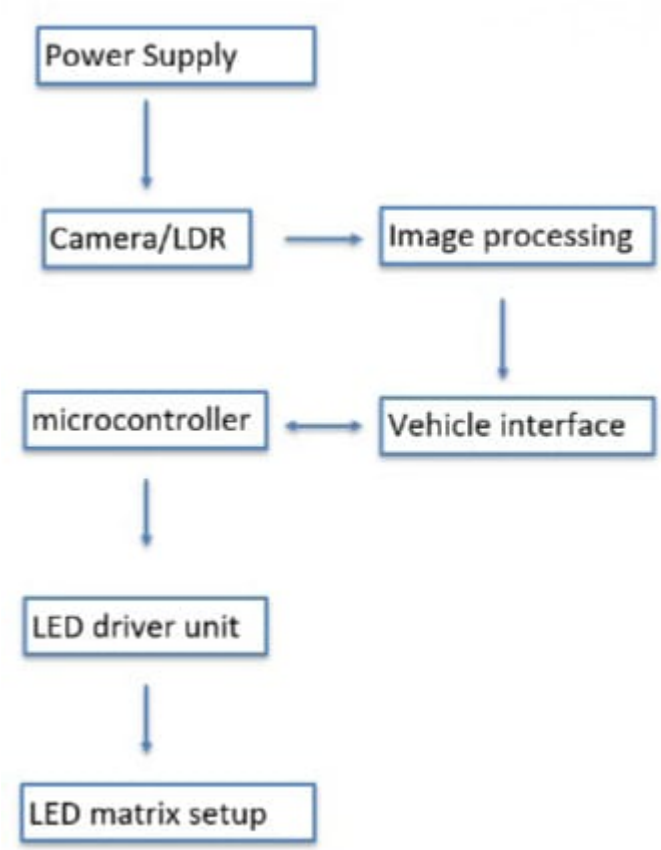


Fig: 4.1

1. Power Supply

Vehicle battery (12V or 24V system) as the input source

Output: Controlled DC power, usually 3.3V, 5V, or 12V, depending on the component.

Function: Gives all components power.

Type: Electrical part (regulator or DC-DC converter)

Function: Provides steady current and voltage

Getting charged Condition: Linked to the car's charging infrastructure; functioning state: System activation trigger or vehicle ignition ON

2. Camera / LDR (Light Dependent Resistor)

Source of Input: Front-road view and ambient light
Output: Image signal or analog/digital light intensity
Function: Captures visual input and/or detects ambient light levels.

Sensor type (CMOS image sensor or LDR)
Function: Provides environmental data for processing
Getting charged Condition: Active when required, powered by a power source
Operation Condition: Insufficient light, detection of approaching traffic

3. Image Processing

Source of Input: Camera/LDR Data
Output: Digital signal with information on object and traffic recognition
Function: Identify vehicles or objects by analyzing light data or images.

Type: Hardware/Software (onboard microprocessor or image processor)
Function: Adaptive lighting decision-making tool
Getting charged Condition: Power supply-operated
Operation State: Processing in real time when the headlamps are turned on

4. Vehicle Interface

Vehicle signals (speed, steering angle, GPS, etc.) serve as the input source.
Output: Information sent to a microcontroller to modify behavior
Function: Contextual awareness by communicating the vehicle's state

Type: Interface for communication (CAN, LIN, Ethernet)

Function: Align lighting with the dynamics of the vehicle

Condition of Charging: Linked to the car's electronic system

When the system is enabled, the operation condition is active.

5. Microcontroller

Data from the vehicle interface and image processing are the input sources.

Output: Signals to the LED driver unit for control

Function: Adaptive lighting control logic

Embedded processor (MCU) type

Function: Making decisions and charging the system controller

Condition: Continuously low power consumption

Operation Condition: Responds to sensor input to operate in real-time.

6. LED Driver Unit

Source of Input: Microcontroller control signals

Output: Controlled current to the matrix of LEDs

Function: Using input commands to control LED power

Circuit type: power electronics

Function: Selectively drives the LED segments

Operating Condition: Vary according to adaptive lighting requirements

Charging Condition: Powered as required during lighting

7. LED Matrix Setup

4.2 Circuit Diagram:

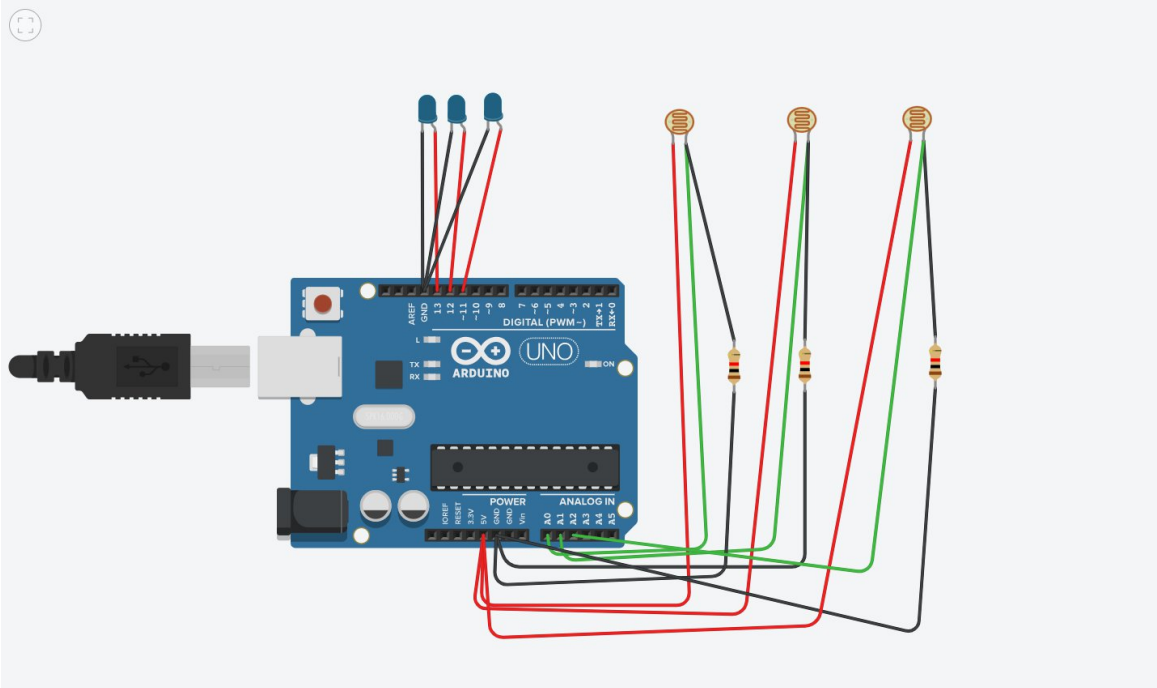


Fig 4.2

1. Arduino UNO

• Type: ATmega328P microcontroller board

• Rating: 5V logic, with a 9V battery or USB power source

• Wiring:

o Attached to the LDR through the analog input pin (A0)

o Uses digital output pins (such as D3, D5, and D6) to control LEDs.

• Function: o Uses the LDR to read ambient light levels.

o Simulates adaptive lighting by controlling the brightness or switching of LEDs based on light intensity.

2. LDR (Light Dependent Resistor)

Analog light sensor type; resistance rating (depending on light): ~10kΩ to ~1MΩ; wiring:

o Connect one terminal to 5V.

o A 10kΩ resistor connected to GND (voltage divider) and the other terminal to Analog Pin A0.

• Its function is to measure ambient light.

o The Arduino modifies the headlamp output and determines the light level by reading the voltage across it.

3. LEDs (x3)

• Type: Bright or white light-emitting diode

• Rating: Usually 20mA, forward voltage between 2V and 3.2V; • Wiring:

o Use 220Ω resistors to connect the anode (long leg) to the digital pins.

o Cathode to GND (short leg).

• Purpose: To replicate the matrix LEDs found in a headlight system.

o To enhance road visibility and prevent blinding others, turn on/off or adjust according to ambient light.

4. Resistors • Wiring: o 10kΩ between GND and LDR

• Function:

o 10kΩ creates a voltage divider with the LDR for analog sensing;

o 220Ω connects each LED to the digital pin.

o To avoid damage, 220Ω resistors restrict the current flowing to LEDs.

5. Connected 9V Battery

• Type: Power source;

• Rating: 9V;

• Wiring:

o Attached to the Arduino power jack;

• Function:

o Supplys power to the Arduino when USB is not in use (for standalone operation in vehicle).

4.3 Key Components and Their Functions;

Component	Function
Microcontroller (e.g., Arduino/STM32)	Controls the system logic and processes sensor inputs.
Camera Module / Light Sensor	Detects incoming traffic, road lighting conditions, and objects ahead.
Matrix LED Headlamp	Composed of multiple LEDs that can be turned on/off independently.
Motor Driver / Servo Motor	Adjusts headlamp angle based on vehicle steering or tilt.
Ambient Light Sensor (LDR)	Detects light levels to decide when headlamps should activate/dim.
Ultrasonic / IR Sensors	Detects proximity of vehicles or obstacles.
CAN Bus / UART Interface	For communication with other vehicle modules.
Power Supply Circuit	Converts vehicle power (12V) to usable voltage levels for components.
Relay Module / MOSFETs	Switches LED segments on/off efficiently.
Temperature Sensor (Optional)	Monitors temperature of the LED system to prevent overheating.

4.4 List of Simulation tools used:

Tool Name	Function / Use Case
MATLAB/Simulink	Simulate control logic, adaptive lighting algorithms, and sensor integration.
Tinker CAD	Simulate electronic circuits, microcontroller (like Arduino/AVR) programming & testing.
AutoCAD Electrical	Design wiring diagrams and electrical schematics of the lighting system.
Arduino IDE	Code and test microcontroller logic for adaptive lighting behavior.

CHAPTER 5

RESULT AND DISCUSSION

5.1 Hardware Setup and Implementation

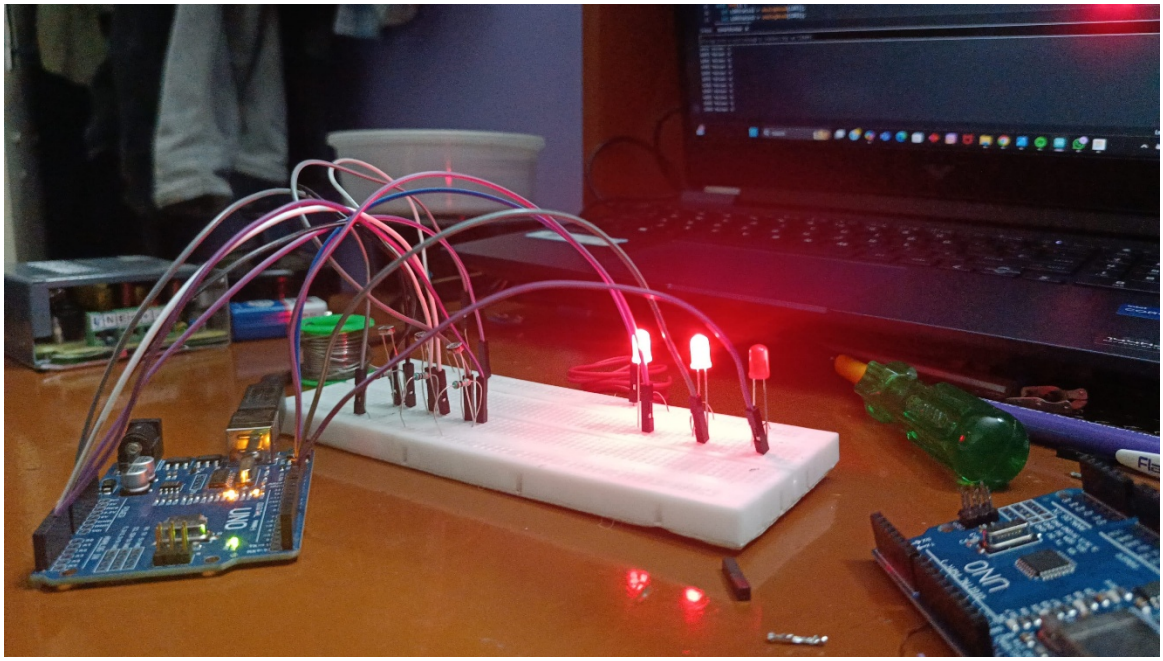


Fig:5.1

5.1.1 Hardware Assembly

To evaluate the Adaptive Matrix Headlamp's viability, a prototype system was put together. The hardware setup consisted of the following parts: a micro-controller Arduino UNO and three independently controllable LEDs

A bread board; light-sensing LDRs; and LDR resistors

As soon as the LDR detects light, the corresponding LED that is connected to it turns off. The other two LEDs are likewise off in a similar manner. which makes driving more comfortable and improves visibility.

5.2 Expected Simulation Behaviors (If Practically Implemented)

5.2.1 LED turn On and OFF Simulation

The simulation software is displayed in Tinker CAD. The three separate LEDs in this are connected to their respective LDRs. The corresponding LED turns off when the LDR detects light. as seen below.

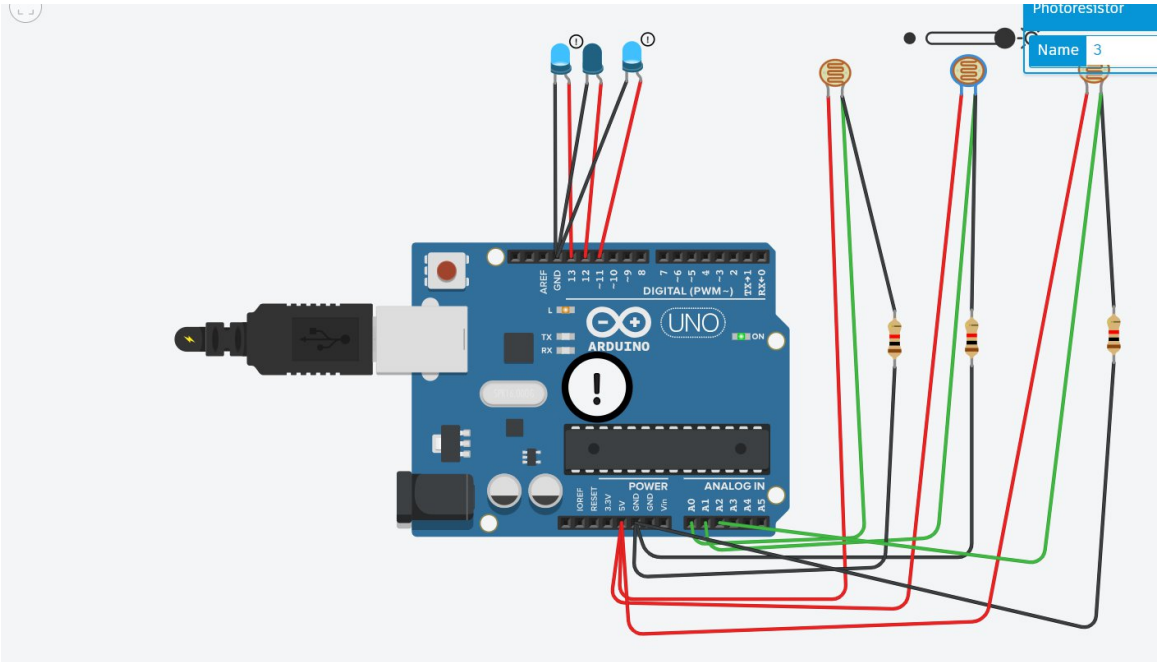


Fig:5.2

5.3Importance of Simulation in Future Work

Practical simulation is not included in the project's current iteration, but adding it in later stages has the following advantages:

• Risk-Free Testing: Without causing physical component damage, circuits can be examined for flaws or inefficiencies.

• Parameter optimization: It is simple and damage-free to test the LED and LDR setups.

• Predictive analysis: Before hardware implementation, future upgrades such as automation and multiple LED controllers can be virtually tested.

• IoT Integration Testing: It is theoretically possible to simulate cloud-based dashboards and remote monitoring (for example, through ThingsBoard or Blynk).

CHAPTER 6

CONCLUSION

An important development in driver assistance and vehicle safety technology is the Adaptive Headlamp Matrix System. This system improves nighttime visibility while reducing glare for oncoming drivers by intelligently modifying light distribution in real time based on environmental factors, road curvature, and traffic conditions. Improved road illumination is possible without endangering the safety of other drivers thanks to its ability to selectively dim particular LED segments. Adaptive headlamp matrix systems are on the verge of becoming a standard feature in contemporary cars as automotive innovation continues to place a high priority on comfort and safety, making driving safer and more effective.

This system's main benefits are as follows:

1. Improved Night Visibility
2. Less glare for approaching vehicles
3. Optimal Lighting Under Every Situation

Vehicle safety is greatly improved by the adaptive headlamp matrix system, particularly when driving at night. It offers the best visibility without blinding other drivers by cleverly adjusting light beams. Drivers can identify road conditions and obstacles sooner thanks to this technology. Additionally, it makes driving safer in a variety of traffic and weather conditions. All things considered, it's a wise and practical move toward safer and more sophisticated car lighting systems.

Bibliography:

1. B. Y. Lee, J. S. Lee, and M. S. Park, "Adaptive Driving Beam System Design and Implementation Using Image Processing," Sensors, vol. 19, no. 9, 2023.

2. "The Impact of Adaptive Headlighting on Nighttime Driving Safety," Lighting Research & Technology, vol. 48, no. 3, pp. 311–323, 2022; M. C. Paul, M. Sivak, and J. Schoettle.

3. "Development of Adaptive Driving Beam Based on LED Matrix Headlamps," SAE Technical Paper 2013-01-0044, 2023, by A. Ishiguro et al.

4. "A Study on Adaptive Driving Beam Control to Improve Visibility Without Glare," SAE Int. J. Passeng. Cars – Mech. Syst., vol. 6, no. 3, 2021, N. Wada et al.

5. "Real-time Control Algorithm for Adaptive Headlight System Using Vehicle Dynamics," International Journal of Automotive Technology, vol. 19, no. 2, 2022, by S. L. Sze and H. C. Yu.

6. Hella KGaA Hueck & Co. was awarded U.S. Patent No. US20120013155A1, "Adaptive Headlight Beam Control System," in 2022.

7. "Matrix Beam Headlamp and Control Method for a Vehicle," European Patent EP2733243A1, Audi AG, 2019.

8. Ronald K. Jurgen, SAE International, 2020, Automotive Lighting and Human Vision.

9. "Adaptive Front-Lighting Systems," by Brian W. Herrick, in Intelligent Vehicle Technologies, CRC Press, 2022

.

10. "Uniform Provisions Concerning the Approval of Adaptive Front-lighting Systems (AFS) for Motor Vehicles," United Nations Economic Commission for Europe, Regulation No. 123, 2020.

11. NHTSA Report: "Effectiveness of Adaptive Headlight Systems in Reducing Crashes," National Highway Traffic Safety Administration, U.S. Department of Transportation, 2020.

12. Kumar, A., Master's Thesis, RWTH Aachen University, 2019, Design of a Smart Headlight System Using Matrix LED Technology.

13. Continental AG, White Paper on Matrix Headlamp Technology, 2021.

14. Valeo Lighting Systems, Safety and Adaptive Lighting: A Technology Briefing, 2020.

15. "The Safety and Performance Advantages of Matrix LED Headlights,"https://www.caranddriver.com/features/matrix-led-headlights-explained/, CarandDriver Retrieved May 20